Fernanda Ventorim

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EDUCATION

Ph.D. in Computer Science **2024-Present**
University of California, Irvine (UCI) *Irvine/CA – USA* | GPA: 4.00/4.00
Advisor: Prof. Nalini Venkatasubramanian

M.Sc. in Computer Science **2022-2024**
University of California, Irvine (UCI) *Irvine/CA – USA* | GPA: 3.81/4.00

M.Sc. in Urban Planning **2018-2020**
Fluminense Federal University (UFF) *Niterói/RJ – Brazil* | GPA: 9.73/10
Advisor: Prof. Vinicius M. Netto

B.Sc. in Architecture and Urban Planning **2012-2016**
Brazilian College (Multivix) *Vitória/ES – Brazil* | GPA: 8.38/10

RESEARCH INTERESTS

Data Science. Human Mobility Modeling. Cyber-Physical Systems. Machine Learning. Smart Communities. Disaster Resilience.

PROFESSIONAL EXPERIENCE

Ph.D. Researcher **2024-Present**
Department of Computer Science, University of California, Irvine
Distributed Systems Middleware Group & Information Systems Research Group

Research Scientist Intern **Summer 2026**
ImageCat – Long Beach/CA
Developed an agent-based tsunami evacuation model by integrating HAZUS flood loss estimates, PEAT accessibility routing, and empirically derived behavioral response models, enabling large-scale simulation of heterogeneous evacuation scenarios.

M.Sc. Researcher **2018-2020**
Department of Urban Planning, Fluminense Federal University, Niteroi/RJ-Brazil
Vinicius M. Netto's Lab

Undergraduate Student Intern **2015-2016**
Petrobras – Vitória/ES-Brazil
Assisted in civil engineering and architectural design projects, contributing to planning, technical documentation, and project management activities.

TEACHING EXPERIENCE

Graduate Teaching Assistant
- Department of Computer Science, University of California, Irvine/CA **2022-Present**
CS121 - Information Retrieval, ICS31 - Introduction to Programming, ICS32 - Programming with Software Libraries, ICS139W - Critical Writing

- Department of Urban Planning, Fluminense Federal University, Niteroi/RJ **2019**
Production and Management of Urban Land Use

Undergraduate Learning Assistant
- Department of Architecture and Urbanism, Brazilian College, Vitoria/ES **2012-2014**
Environmental Comfort I, Visual Arts

RESEARCH PROJECTS

Graduate Student Researcher – UCI **2024-Present**
Project: "NSF-JST SHIELD: Enabling human-centered digital twins for community resilience"
PI: Prof. Nalini Venkatasubramanian

- Developing **PARA-FLOW**, a framework for accessibility-aware multi-modal mobility modeling under routine and disaster conditions.
- Integrating behavioral modeling, geospatial analysis, and transportation network constraints to generate realistic human mobility trajectories for vulnerable populations.

Graduate Student Researcher – UCI **2024-Present**

Project: "CareDEX: Enabling disaster resilience in aging communities via a secure data exchange"

PI: Prof. Nalini Venkatasubramanian

- Contributed to the development of the **RADAR** framework, an accessibility-aware mobility model to estimate evacuation demand and transportation needs for older adults during disasters.
- Contributed to the design of geospatial decision-support tools for emergency resource allocation and community resilience.

Master's Thesis – UFF **2018-2020**

Project: "Criminality and urban space: an analysis of the relationship networks between types of crime, victim and spatial location in Rio de Janeiro"

Supervisor: Prof. Vinicius M. Netto

- Developed a complex-network framework to uncover non-random associations among crime types, victim profiles, and urban locations using large-scale crime data from Rio de Janeiro.
- Implemented graph-based algorithms and statistical analyses, revealing crime patterns and demographic disparities among victims.

PUBLICATIONS

Manuscripts Under Review

[UR-1] **F. Ventorim**; A. Chio; R. Vescovo; E. Mas; S. Koshimura; N. Venkatasubramanian. 2026. PARA-FLOW: An Event-Based Framework for Modeling Accessible Multi-Modal Urban Mobility.

Journal Papers

[J-2] **F. Ventorim**; V. M. Netto. 2024. The Hidden Connections of Urban Crime: A Network Analysis of Victims, Crime Types, and Locations in Rio de Janeiro. *Urban Science*, 8(2):72. https://doi.org/10.3390/urbansci8020072

[J-1] **F. Ventorim**; V. M. Netto. 2023. Crime and urban space: the relationship networks between crime, victims, and location in Rio de Janeiro. *urbe. Revista Brasileira de Gestão Urbana*, *15*, p.e20220141. https://doi.org/10.1590/2175-3369.015.e20220141

Conference Proceedings

[C-4] M. M. Kenne; **F. Ventorim**; C. Cossey; Z. Hu; J. Rousseau; N. Venkatasubramanian. 2025. *RADAR: Resource Allocation for Disaster Resilience in Senior Health Care.* In: Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL'25), 1–14.

[C-3] **F. Ventorim**; F. P. Dias; J. T. Araujo; R. M. Correa. 2019. *Contemporary city and appropriation of public space: notes on a sensitive analysis of public spaces in the Botafogo neighborhood in Rio de Janeiro/RJ.* In: Proceedings of the 8th Conferência da Rede Lusófona de Morfologia Urbana – PNUM, Maringá, 879-895.

[C-2] **F. Ventorim**; R. M. Lora. 2017. *Connected urbanity: a reconciliation between urban voids and open spaces in the Jesus de Nazareth slum.* In: Proceedings of the 6th Conferência da Rede Lusófona de Morfologia Urbana – PNUM, Vitória, 293-301.

[C-1] **F. Ventorim**; L. R. Bragatto; M. L. Rodrigues; I. P. Dias; A. C. Marques; A. C. Diniz; G. Achiamé; A. S. Sauer; F. Rembiski. 2014. *Analysis of natural lighting performance in drawing classrooms in Vitória (ES).* In: Proceedings of the 15th Encontro Nacional de Tecnologia do Ambiente Construído – ENTAC, Maceió, 897.

News and Media Mentions

[N-1] Alana Gandra. *CNN Brazil*.
https://www.cnnbrasil.com.br/nacional/mulheres-sao-maiores-alvos- da-criminalidade-no-rio-de-janeiro-diz-estudo/

AWARDS AND SCHOLARSHIPS

M.Sc. Scholarship: Full M.Sc. funding scholarship from the Brazilian research agency CAPES (2018).

Undergraduate Research Scholarship: Scholarship from the Brazilian research agency FAPES (2014).

ACADEMIC ACTIVITIES

Assistant Editor for Urban Morphology Journal – ISSN 2182-7214 (2019-2021).

Graduate Student Representative (2018).

Certificate degree: Teaching and Management in Higher Education (2016-2017) - Research: *Between practice and teaching in Architecture and Urbanism* | GPA: 9.67/10.

SKILLS

Programming & Tools: Python, SQL, HTML, LaTeX, Jupyter Notebooks, Git

Software & Tools: ArcGIS, Stata, SPSS, AWS, Linux, Grafana, Figma, Jupyter Notebooks, Git